

Single-Channel RTD Interface

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Reference: Octavia_Carbon_Schematic.pdf

GitHub: https://github.com/KeenanKE/Octavia_Carbon

I. Design Description

This design accommodates both Pt100 and Pt1000 sensors using a 4-wire configuration. As a fall-back, a pair of SPDT switches (SW1 and SW2) enables manual selection of the active RTD type.

- SW1 selects which RTD receives the excitation current from the LM134H programmable current source.
- SW2 selects the corresponding voltage sense line going into the INA333 instrumentation amplifier.

II. Noise Mitigation and Minimization of Self-Heating

To minimize self-heating, the LM134H is configured with an external resistor (R1) to produce a low excitation current ($\sim 250 \mu\text{A}$), which reduces power dissipation in the RTD element.

To mitigate electrical noise, the design includes:

- A low-pass RC filter (R3 and C1) between the INA333 output and the ADS1118 ADC input to attenuate high-frequency interference.
- TVS protection diodes (D1, D2, D3) at the INA333 input terminals to suppress transient spikes and ESD events.
- Short, shielded signal paths and proper grounding to minimize electromagnetic interference from the ESP32 or surrounding digital systems.

III. Reliability at Elevated Temperatures (up to 101 °C)

All selected components are rated for operation beyond 101 °C:

INA333, ADS1118, LM134H, and passive components are specified for $\geq 105 \text{ °C}$.

The OLED display, though not essential to core functionality, is selected with a conservative placement away from heat sources.

The PCB layout (not generated) should ensure thermal isolation of sensitive analog sections from the ESP32's Wi-Fi transceiver and step-down regulator (LM2596). The system should therefore be electrically and thermally stable in harsh DAC operating conditions.

IV. Firmware Implementation Overview

a. SPI Initialization

The ESP32 initializes the SPI interface using its VSPI bus, with connections to the ADS1118 ADC (DIN, DOUT, SCLK, CS) as per the schematic. The SPI is configured for mode 1, and a standard 16-bit data exchange is used to communicate with the ADS1118.

b. Reading Measurement Data

The firmware performs periodic SPI transactions to:

- Initiate a single-shot conversion on the ADS1118
- Poll the DRDY flag
- Read back the raw ADC value (AIN0 input corresponding to RTD voltage drop)

c. Temperature Conversion Logic

The raw ADC value is converted into voltage, then into resistance using the known excitation current (from LM134H), and finally into temperature using the Callendar–Van Dusen equation:

$$R(T) = R_0(1 + \alpha T), \text{ solved for } T$$

Separate scaling logic is implemented for Pt100 (100 Ω at 0 °C) and Pt1000 (1000 Ω at 0 °C) using a configuration flag.

V. RTD Type Selection

Although RTD selection is hardware-based via switches (SW1, SW2), the firmware includes a manual toggle option in software to adjust expected resistance range and lookup constants and conversion equations. This satisfies the bonus requirement by enabling runtime switching logic that adjusts scaling automatically based on user selection or GPIO input.